

Data-Driven Optimization of Lattice Structures

ME-438 – Data-Driven Design and Fabrication Methods

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1 Introduction

Lattice structures are increasingly used in lightweight engineering because they can achieve high stiffness-to-weight and strength-to-weight ratios by distributing material along load-carrying struts rather than filling a solid volume. This makes them particularly attractive for applications where mass reduction is critical, such as aerospace brackets, energy-absorbing components and patient-specific implants.

However, designing an efficient lattice is not straightforward. Its mechanical performance depends on coupled geometric and topological parameters, including strut radius, cell subdivision and unit-cell connectivity. These parameters create a high-dimensional mixed design space, combining continuous, integer and categorical variables. In addition, the structural response is difficult to predict analytically, especially because local stress concentrations may appear at strut junctions.

Since each candidate design requires a full three-dimensional finite element simulation, exhaustive exploration of the design space would be computationally expensive. This motivates the use of data-driven optimization methods, which can learn from a limited number of high-fidelity simulations and guide the search toward promising regions of the design space.

In this work, a parametric lattice generation and FEM pipeline is combined with Gaussian Process surrogate models and the NSGA-II multi-objective optimization algorithm. The objective is to identify the Pareto-optimal trade-off between structural mass and mechanical resistance while keeping the number of expensive simulations limited.

2 Design Problem

2.1 Problem Formulation

This study considers a 50 mm cubic lattice structure made of PLA ($E = 2.2$ GPa, $\nu = 0.4$, $\rho = 1270$ kg/m³). The structure is subjected to a compressive load, modelled as a uniform pressure applied on the top face while the bottom face is fully clamped. This configuration provides a simple and

reproducible test case for comparing lattice designs, while remaining representative of lightweight structural components loaded in compression.

The optimization problem is formulated as a two-objective minimisation of the structural mass $m(x)$ and peak von Mises stress $\sigma_{\text{VM}}^{\text{max}}(x)$, used as a proxy for structural integrity:

$$\min_{x \in \mathcal{X}} F(x) = (m(x), \sigma_{\text{VM}}^{\text{max}}(x)). \quad (1)$$

These two objectives are conflicting: reducing mass generally decreases material usage and print time, but may increase stress and reduce the safety margin. No scalarisation is therefore imposed before the optimization. Instead, the full Pareto front $\mathcal{P}^*$ is sought, allowing the final design to be selected afterwards using an explicit preference criterion.

2.2 Design Variables and Space Reduction

The complete lattice design space is high-dimensional and mixed, since strut radii, cell subdivisions, junction reinforcements and unit-cell connectivity could all be varied independently. As exhaustive exploration with high-fidelity FEM simulations was not tractable, the space was reduced to three variables based on preliminary tests of simulation cost and numerical robustness. The retained variables are summarised in Table 1.

Table 1: Retained design space after preliminary exploration.

Var.	Description	Type	Domain
r	Radius	Cont.	$[1.0, \min(2.0, 0.45a/n)]$ mm
n	Cells/axis	Int.	$\{1, 2, 3\}$
t	Topology	Cat.	$ \mathcal{T} = 7$

Topology. The unit cell initially allowed $2^4 = 16$ combinations of four diagonal families: face diagonals in the XY, XZ and YZ planes, and body diagonals. XY face diagonals, perpendicular to the compression axis, were excluded because they mainly add mass without improving axial load transfer. Voronoi and Delaunay stochastic topologies were also tested but discarded due to recurring mesh generation failures. The seven retained configurations are shown in Figure 1.

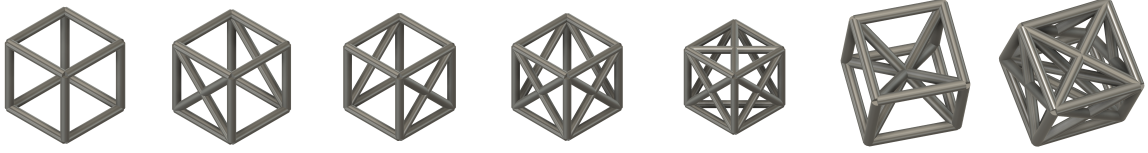


Figure 1: The seven retained unit cell topologies, shown for $n = 1$. From left to right: none, XZ, YZ, XZ+YZ, XY+XZ+YZ, body, and full.

Strut radius and cell count. The strut radius was constrained to avoid overlap between adjacent struts:

$$r \leq 0.45 \frac{a}{n}, \quad a = 50 \text{ mm}. \quad (2)$$

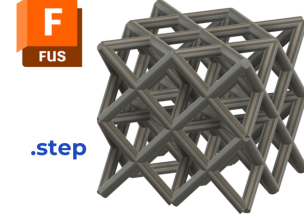
A lower bound of $r \geq 1.0 \text{ mm}$ was imposed because smaller radii led to degenerate tetrahedral meshes and unreliable stress fields at strut junctions. Similarly, $r \leq 2.0 \text{ mm}$ and $n \in \{1, 2, 3\}$ were retained to keep simulation times compatible with a reasonable DoE size. These ranges therefore reflect both geometric feasibility and practical data-generation constraints.

3 Experimental Setup and Data Pipeline

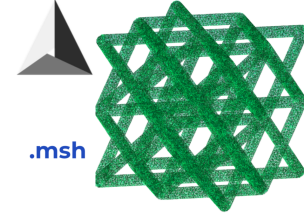
3.1 Simulation Pipeline

Each design evaluation follows a three-stage Python pipeline that automatically transforms a set of design variables into simulation-ready data:

1. **Geometry generation:** the lattice geometry is generated parametrically from the design variables: strut radius, number of cells per axis and unit-cell topology. It is built in **gmsh** using OpenCASCADE cylindrical primitives, fused through Boolean union operations, and exported as a **.step** file.
2. **Meshing:** the CAD geometry is automatically converted into a tetrahedral finite element mesh using **gmsh**. The mesh size is controlled by a single parameter, allowing a compromise between numerical accuracy and computational cost. This step produces a **.msh** file that can be directly imported into the FEM solver.
3. **FEM solve:** the mesh is imported into DOLFINx, where the linear elastic compression problem is solved. Material properties, boundary conditions and loading are assigned consistently for all designs: the bottom face is fixed and the compressive load is applied on the top face. The solver outputs displacement, von Mises stress σ_{VM} and safety factor fields as **.pvd** files for post-processing.



(a) Input parameterisation of the lattice model, including geometry, topology, material, mesh and load.



(b) Automatic mesh generation in **gmsh**, producing a tetrahedral mesh for finite element analysis.



(c) Finite element simulation output, post-processed to obtain displacement, von Mises stress and safety factor.

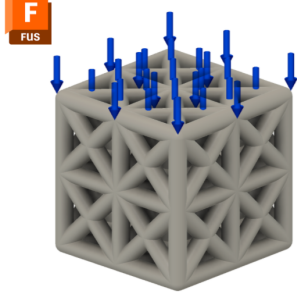
Figure 2: Automated data-generation pipeline used for each lattice design, from parameter definition to meshing and FEM post-processing. The workflow produces successive **.step**, **.msh** and **.pvd** files.

3.2 Model Verification

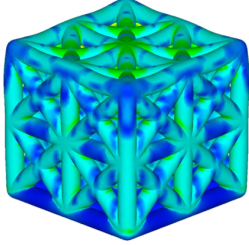
Before generating the optimization dataset, the Python-based FEM pipeline implemented with DOLFINx was verified against a reference Fusion simulation. The goal was to confirm that the compression case, boundary conditions, load application and post-processing were correctly reproduced, since the surrogate models and subsequent optimization rely entirely on generated data.

Figure 3a shows the reference model with the compressive load applied on the top surface and the bottom surface fully fixed. Figure 3b shows the corresponding complete simulation result, used as a stable reference for validating the Python workflow.

The comparison between the Fusion reference and the DOLFINx simulation showed an error below 7%. This level of agreement was considered sufficient to validate the pipeline for the optimization study. It confirms that the generated meshes, boundary conditions and stress post-processing are consistent enough to support the data-driven process.



(a) Fusion simulation model used as reference, showing the applied compression load case and fixed bottom surface.



(b) Complete Fusion simulation result used to verify the consistency of the Python DOLFINx pipeline.

Figure 3: Model verification through comparison between the Python DOLFINx pipeline and a reference Fusion simulation.

3.3 Data Quality and Cost

Each FEM evaluation requires approximately 2–5 minutes on a single CPU core, depending mainly on mesh density and lattice complexity. This cost remains acceptable for a limited design of experiments, but makes exhaustive exploration of the full design space impractical. It therefore motivates the use of surrogate-assisted optimization after an initial simulation dataset has been generated.

The main data-quality issue is the presence of local stress concentrations at strut junctions, as visible in Figure 3b. Since the lattice is modelled as fused cylindrical struts, sharp intersections may generate artificially high peak von Mises stresses, especially with relatively coarse meshes. These local peaks may not represent the global structural behaviour and could bias the optimization if only the raw maximum stress were used. For this reason, both maximum and percentile-based stress criteria are considered, with their influence on the Pareto front discussed in Section 5.4.

4 Data-Driven Optimization Methodology

4.1 Method Selection

The design problem is a black-box optimization task: the objective map $F : \mathcal{X} \rightarrow \mathbb{R}^2$ is accessible only through the FEM pipeline described in Section 3.1, and its gradient is unavailable. The design space also combines continuous, integer and categorical variables. Finally, mass and stress are not directly commensurable, so imposing a scalar objective before the search would introduce an arbitrary preference.

These properties motivate a two-stage methodology. First, Gaussian Process (GP) surrogates are fitted to the high-fidelity simulations to approximate both objectives with inexpensive probabilistic models. Second, NSGA-II explores the surrogate landscape to recover the Pareto front $\hat{\mathcal{P}}^*$. This combination is appropriate because GPs work well with limited data, while NSGA-II handles mixed-variable multi-objective optimization without gradient information.

4.2 Design of Experiments

The training set is constructed by Latin Hypercube Sampling (LHS), chosen for its space-filling properties under a limited budget. Since n only takes three admissible values, the sampling is stratified across $n \in \{1, 2, 3\}$, and a one-dimensional LHS is drawn over $r \in [1.0, 2.0]$ mm within each stratum. Samples violating the feasibility condition (2) are projected onto the feasible bound.

The procedure is repeated independently for each of the seven topologies, yielding $N_{\text{DoE}} = 16$ samples per topology and $7 \times 16 = 112$ candidate designs in total. Each design is evaluated by the full FEM pipeline to populate the dataset $\mathcal{D} = \{(x_i, F(x_i))\}_{i=1}^{112}$.

4.3 Surrogate Setup and Objective Function Design

Architecture. For each topology $\tau \in \mathcal{T}$, two independent GP regressors are fitted on $\mathcal{D}_\tau$: one for mass and one for stress. The inputs are the strut radius r and the cell count n , with n treated as a discrete numerical variable. The categorical topology is not encoded directly; instead, one GP pair is trained per topology, avoiding an artificial distance between topologies and reflecting their discontinuous effect on load paths.

Stress criterion tuning. The raw maximum stress $\sigma_{\text{VM}}^{\text{max}}$ is strongly affected by numerical stress concentrations at strut junctions. Preliminary DoE results revealed extreme outliers exceeding 2000 MPa at low mass, corresponding to near-singular mesh configurations rather than physical failure. The 97th-percentile stress σ_{VM}^{P97} was therefore adopted as the main stress criterion, filtering out localised numerical artefacts while preserving sensitivity to genuine stress variations. A comparison with σ_{VM}^{P95} confirmed that the front shape and topology dominance were robust to this choice.

Log-transform and preprocessing. The stress response is modelled in logarithmic space, $\log(\sigma_{\text{VM}})$, to reduce the influence of high-stress points and guarantee positive stress predictions after exponentiation. Mass is modelled directly. Input variables are standardised, and kernel hyperparameters are estimated by maximising the log-marginal likelihood with several random restarts. The implementation uses `scikit-learn`'s `GaussianProcessRegressor`.

4.4 Multi-Objective Search

The trained surrogates are used as cheap substitutes for the FEM solver. The search is performed on the predicted objective values:

$$\min_{x \in \mathcal{X}} [\hat{m}(x), \hat{\sigma}_{\text{crit}}(x)], \quad (3)$$

where $\hat{m}(x)$ and $\hat{\sigma}_{\text{crit}}(x)$ are the GP-predicted objectives, with stress exponentiated after prediction. Constraint (2) is enforced during the search.

NSGA-II is run with a population of 80 individuals over 100 generations, yielding 8000 surrogate evaluations. This fixed evaluation budget was used as the stopping criterion and was selected from preliminary convergence checks, where the Pareto front showed negligible improvement beyond 100 generations. No separate dummy-data benchmark was used; these preliminary checks instead guided the tuning of the optimization parameters. The implementation uses the `pymoo` library with its `MixedVariableGA` interface.

4.5 Surrogate Validation

Before the multi-objective search, the fitted surrogates were checked on the topology-specific datasets. The mass models reached $R^2_{\text{mass}} = 1.000$ for all topologies, which is consistent with mass being a mostly deterministic geometric quantity. The stress models were less direct but still showed good agreement, with R^2_{stress} values between 0.982 and 1.000 after filtering two numerical outliers.

These R^2 values were computed in-sample and therefore measure how well each GP reproduces its own DoE points rather than its true predictive generalisation. They were therefore complemented by a more relevant validation step: representative surrogate Pareto candidates were re-evaluated with the full FEM pipeline and compared with the predicted front. This ensured that the optimization was not driven by surrogate artefacts.

4.6 Verification and Design Selection

A subset of approximately ten representative designs is selected along $\hat{\mathcal{P}}^*$ and re-evaluated using the full FEM pipeline. For each verified design, the surrogate error is quantified as:

$$\Delta_m = \frac{\hat{m}(x) - m_{\text{FE}}(x)}{m_{\text{FE}}(x)}, \quad (4)$$

$$\Delta_\sigma = \frac{\hat{\sigma}_{\text{crit}}(x) - \sigma_{\text{crit,FE}}(x)}{\sigma_{\text{crit,FE}}(x)}.$$

Once the verified front is obtained, a single fabrication candidate is selected by normalising mass and stress over the verified Pareto set and minimising:

$$J(x) = 0.75 \tilde{m}(x) + 0.25 \tilde{\sigma}(x). \quad (5)$$

The weight of 0.75 on mass reflects the primary goal of reducing material consumption and print time, since all Pareto candidates already satisfy the stress constraint with a safety margin.

5 Results

5.1 DoE Exploration

Figure 4 shows the mass and maximum von Mises stress obtained for the 112 designs of the initial DoE. These sampled designs provide the baseline against which the surrogate-assisted Pareto search is compared. The mass response is regular: it increases with the strut radius r , and richer topologies are heavier because they contain additional strut families.

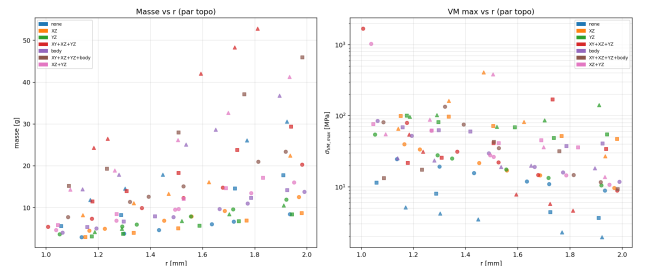


Figure 4: DoE exploration results showing the influence of the strut radius r on lattice mass and maximum von Mises stress at 25 kg load case.

In contrast, the stress response is less direct: no clear monotonic trend appears between $\sigma_{VM}^{\max}$ and r , and similar radii can produce different stress levels depending on topology and cell subdivision. This confirms that the optimization cannot be reduced to a single geometric parameter. The DoE also highlights extreme low-mass stress values, showing the sensitivity of the raw maximum stress criterion to localised numerical peaks.

5.2 Pareto Front

The surrogate-based multi-topology Pareto front for the 25 kg compression case is shown in Figure 5. The **none** topology dominates the useful mass range, approximately between 2 and 33 g, and provides the best mass–stress trade-off.

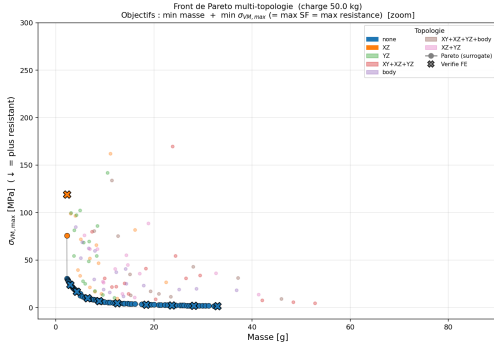


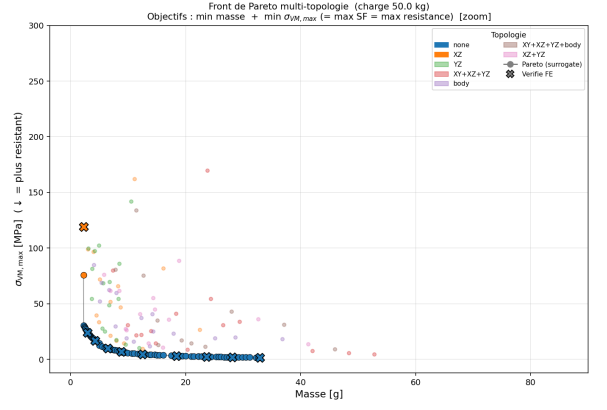
Figure 5: Pareto front obtained for the 25 kg compression load case.

Richer topologies mainly appear at higher mass without sufficient stress reduction, meaning that diagonal and body struts add material without improving load-bearing efficiency enough under uniaxial compression. The FEM verification points remain close to the surrogate front, confirming that the dominance of the **none** topology is not a surrogate artefact.

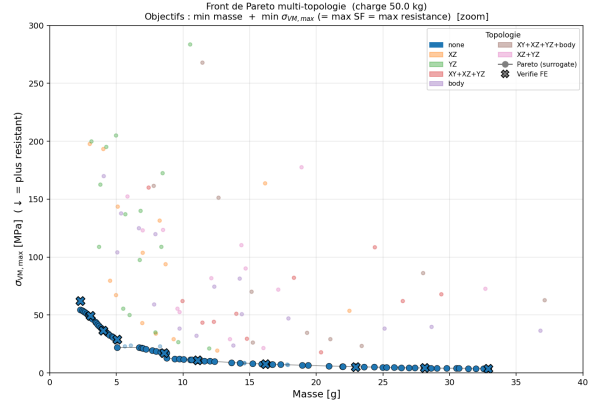
As a benchmark, the Pareto front was compared against the initial DoE samples and all retained topology families. The optimization refines the sampled mass–stress trade-off and shows that richer topologies remain dominated over the relevant mass range.

5.3 Robustness to Load Magnitude

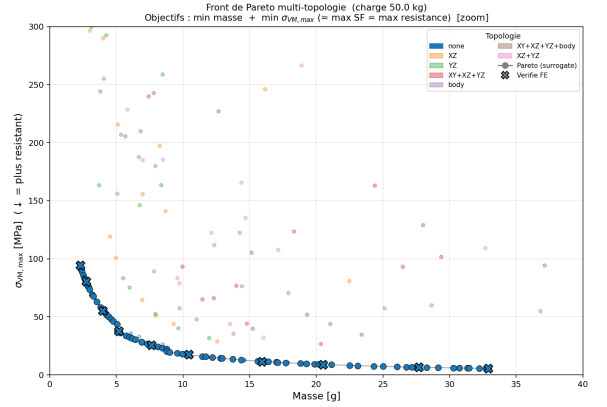
Figure 6 compares the Pareto fronts obtained for 25, 50 and 75 kg. As expected for a linear elastic model, increasing the load shifts the stress levels upward almost proportionally, while the relative organisation of the fronts remains similar. In all cases, the **none** topology dominates the useful mass range, showing that the optimal design family is robust over the investigated load interval.



(a) 25 kg.



(b) 50 kg.



(c) 75 kg.

Figure 6: Comparison of Pareto fronts for the three compression load cases.

5.4 Stress Criterion Comparison

Figure 7 compares the 50 kg Pareto fronts obtained with $\sigma_{\max}$, σ_{P97} and σ_{P95} . Percentile-based criteria reduce the influence of isolated stress peaks and local stress concentrations, but the overall front structure remains stable. The **none** topology still dominates the useful mass range, confirming that the main conclusion does not depend on the exact stress metric. Although σ_{P95} makes some heavier topologies more competitive in stress, they remain penalised by their higher mass and are not selected under the mass-oriented preference.

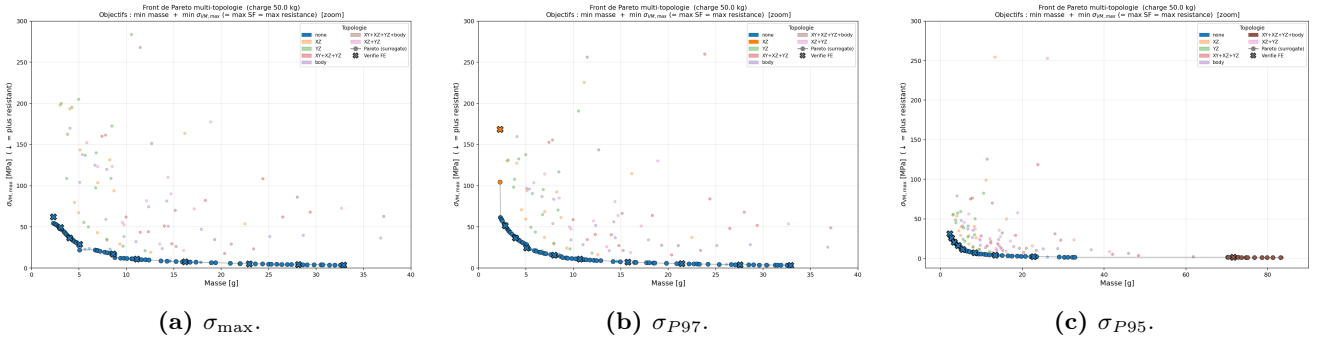


Figure 7: Comparison of Pareto fronts obtained with different stress criteria at 50 kg.

5.5 Optimal Design

The final selection is reported for the 25 kg reference load case, while the 50 kg case was used to assess the robustness of the stress criterion. The final design was selected from the verified Pareto candidates using the weighted score. For the 25 kg load case with the σ_{P95} criterion, the selected optimum is :

$$r = 1.264 \text{ mm}, \quad n = 2, \quad t = \text{none}.$$

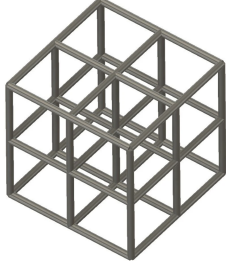


Figure 8: Selected optimal lattice design for the 25 kg load case with $r = 1.264$ mm, $n = 2$ and **none** topology.

Its corresponding performance is:

$$m = 7.98 \text{ g}, \quad \sigma_{VM} = 7.84 \text{ MPa}, \quad SF = 3.82.$$

This design provides the best compromise, although it is neither the lightest nor the lowest-stress solution. Its **none** topology is consistent with the Pareto analysis, which showed dominance across load and stress criteria.

6 Discussion and Conclusion

6.1 Method Performance

The GP+NSGA-II pipeline recovered the Pareto front from only 112 FEM evaluations in approximately 2 hours, while replacing thousands of costly FEM runs by inexpensive surrogate evaluations. Representative Pareto candidates were validated by FEM spot-checks. The dominance of the **none** topology is counter-intuitive but mechanically consistent: under uniaxial compression, orthogonal struts carry load directly, while diagonals mainly add mass.

6.2 Limitations and Future Work

- **Stress concentrations:** junctions are modelled as perfectly bonded cylinders, which can overestimate local stresses. Adding fillets as design variables would better represent lattices.
- **Surrogate accuracy:** 16 points per topology may be insufficient in highly nonlinear regions. Adaptive sampling, such as EHVI, could refine the Pareto front more efficiently.
- **Physical validation:** compression tests would provide direct validation of the FEM results and reveal the effect of assumptions such as perfect bonding, homogeneous material properties and absence of printing defects.
- **Extended design and loading space:** Voronoi and Delaunay lattices were excluded due to meshing issues, and only uniaxial compression was studied. Future work could include stochastic geometries and combined loading cases, where diagonal reinforcements may become more efficient.

6.3 Conclusion

This work demonstrated a data-driven pipeline for multi-objective lattice optimization, from parametric geometry generation to surrogate-assisted Pareto search. With only 112 high-fidelity FEM evaluations, the GP+NSGA-II approach recovered the Pareto front and efficiently explored a mixed design space with continuous, integer and categorical variables.

The main finding is that simple orthogonal lattices without diagonals are Pareto-optimal under uniaxial compression. Although counter-intuitive, this result is mechanically consistent and robust to load magnitude from 25 to 75 kg and to the stress metric used. Overall, the study highlights the value of data-driven optimization for lightweight structural design under limited simulation budgets.